



MOONSHOT

The millions of TV, laptop and smartphone screens across India that are usually airing news, entertainment or cricket matches, were unexpectedly tuned in to watch the launch of the Chandrayaan-3 lunar mission on July 14. Meanwhile, in restaurants across Delhi NCR, patrons and servers alike broke into loud applause to celebrate the moment space agency Indian Space Research Organisation's heaviest rocket LVM3 successfully lifted off from the iconic spaceport at Sriharikota.

Described as a follow-on mission to Chandrayaan-2, it seeks to demonstrate the country's end-to-end capability in safe landing and roving on the lunar surface. The propulsion module will bear the lander and the rover up to 100 km of the lunar orbit. The probe will perform experiments to measure the thermal conductivity and temperature, seismicity, plasma density and elemental composition around the landing site on the moon's South Pole, which has

barely been covered by earlier missions. A passive laser retroreflector array from Nasa is also included in the payload and will be used in lunar laser ranging studies.

Taking lessons from the failure of the Chandrayaan-2, Isro has this time performed rigorous soft-landing tests on an artificial lunar surface located in Ullarthi Kaval village in Karnataka's Chitradurga district. The superfine moon soil was created by crushing anorthosite rocks sourced from Tamil Nadu. The mission is expected to touch down at the South Pole on August 23.

A successful mission will make India the fourth nation after the US, China and Russia to have safely landed and roved on the lunar surface. At ₹615 crore—a little over \$75 million—Isro has again developed a mission on a shoe-string budget. As Isro Chief Sridhar Panicker Somanath said, "Congratulations India! Chandrayaan-3 has started its journey towards Moon."

— Manish Pant

decided to open up the space sector to the private sector. Kaul, who is critical of the way the old Bill was framed, believes that any draft legislation should be thoroughly thought through, as space exploration is already a highly regulated industry. "When companies that have been suppliers to Isro since 1972 are governed under the normative laws of the country, subject to special terms of the contract, sectoral guidelines and so forth, how could it possibly have been contemplated by the drafters that the new space companies would be governed under a different, rather burdensome, legal regime and law? Are the existing players going to be happy to be under this?"

As Indian businesses globalise, local laws may need to match international norms. For instance, international aircraft lessors cannot immediately recover their assets if an airline in India goes bankrupt since India has not joined the Cape Town Convention and Protocol (CTCP). Start-up founders see the implementation of the Bill as complementing the space policy. "An early clearance of the Space Activity Bill will complement the Space Law but also open up the scope for unlimited innovation-led home-grown start-ups to be launched," says Suyash Singh, CEO & Co-founder of Bengaluru-based satellite maker GalaxEye Space. Arindrajit Chowdhury, CEO of space factory start-up Inspecity, also believes that an early passage of the Bill will benefit the fledgeling industry in many ways. "The Bill will pave the way forward for any actor in the space community to develop technologies with the proper authorisation and frameworks, without being concerned about the legalities of the action!"

Besides, Deloitte's Varghese believes a space activity law will provide a solid foundation for more companies entering the sector. "Innovation thrives when companies and entities feel protected, and understand risks